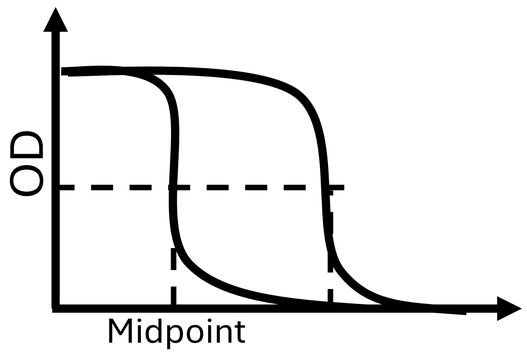
Quantifying the Longevity of Antibody Responses: Progress and Problems with Power Law Decay

Cora Hirst, Hasan Ahmed, Ananya Saha, Andreas Handel, Rustom Antia

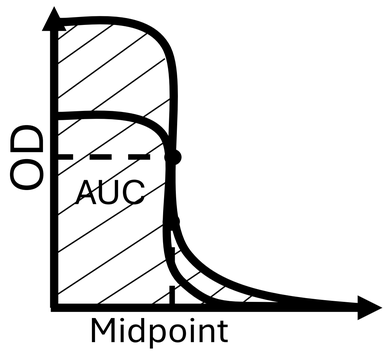
EMORY

Measuring Titers: AUC vs midpoint

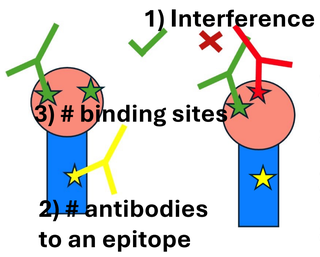
Use midpoint when measuring waning in an individual over time.



Use AUC when comparing titers in different individuals



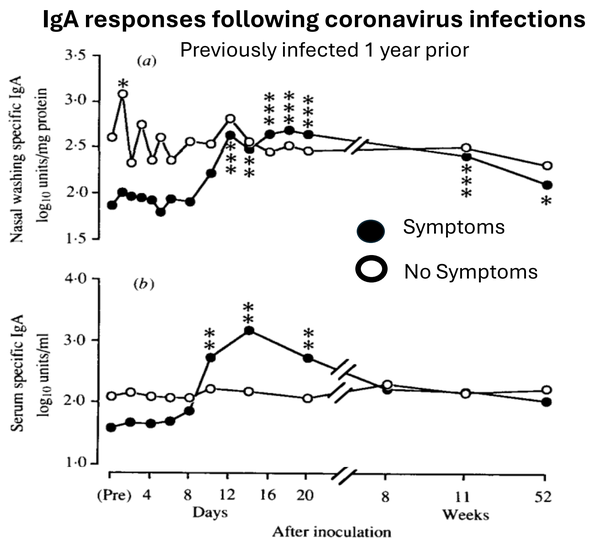
Antibody titer measurements are complicated by the stoichiometry and geometry of antibody binding.

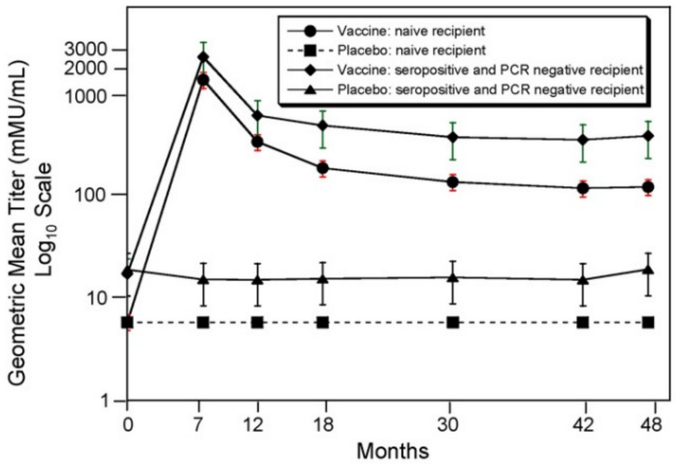


1) Interference decreases with dilution or time.
2) Antibodies to immunogenic epitopes stay saturated at greater dilutions.
3) A larger number of binding sites on an antigen means an antibody to that epitope loses saturation faster.

Open challenges

Do antibodies of all isotypes and in all physiological compartments wane with the same dynamics? Should we expect this? (from Callow et al, Epidemiol Infect, 1990)

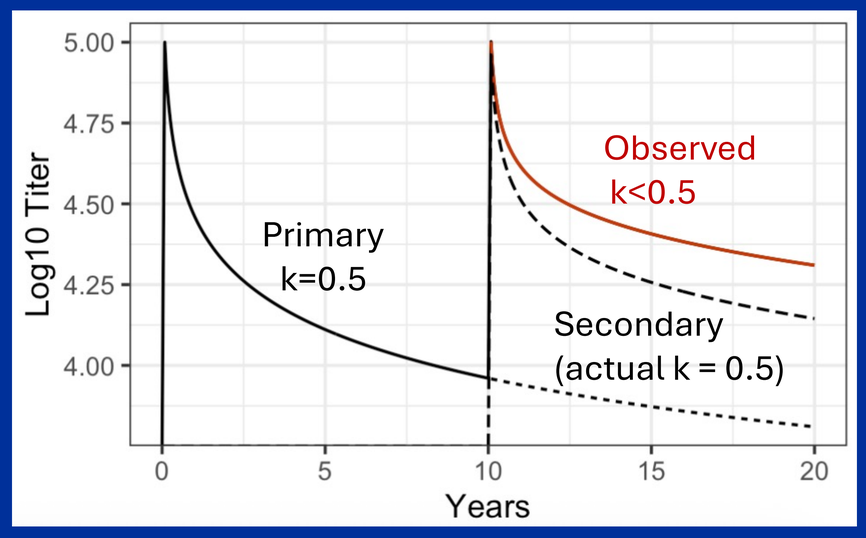


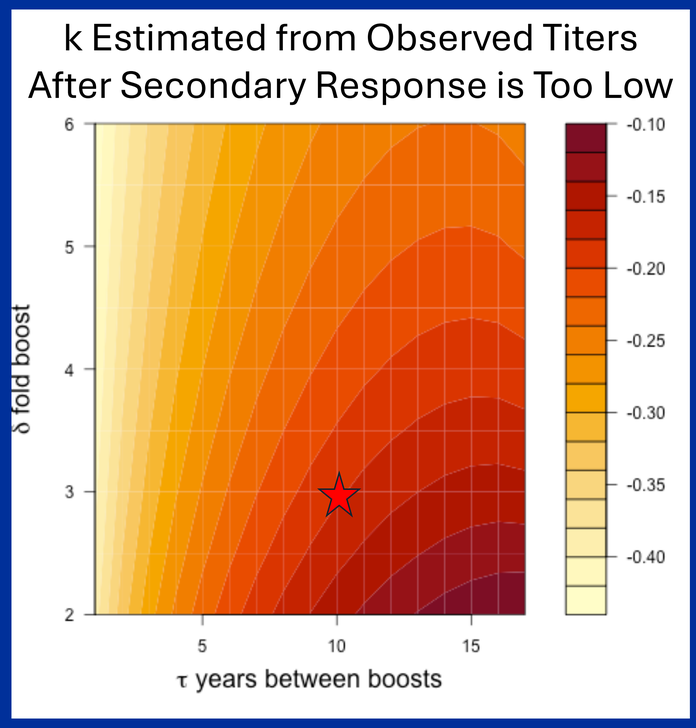


The waning of antibodies to the HPV vaccine may not follow a power law (from Fraser et al, Vaccine, 2007)

Accounting for Pre-existing Immunity During **Analysis**

Adding power-laws does not result in a a power law





As the time between infections τ increases, or the the fold boost of the secondary response δ decreases, the estimated power law coefficient *k* decreases from the true, expected value (of 0.5 in these simulations).

Residual titers from the primary response decrease more slowly than newly generated titers from secondary response – and what we observe is the sum of both.

Secondary responses may seem to decay more slowly, when they actually follow the same power law.